

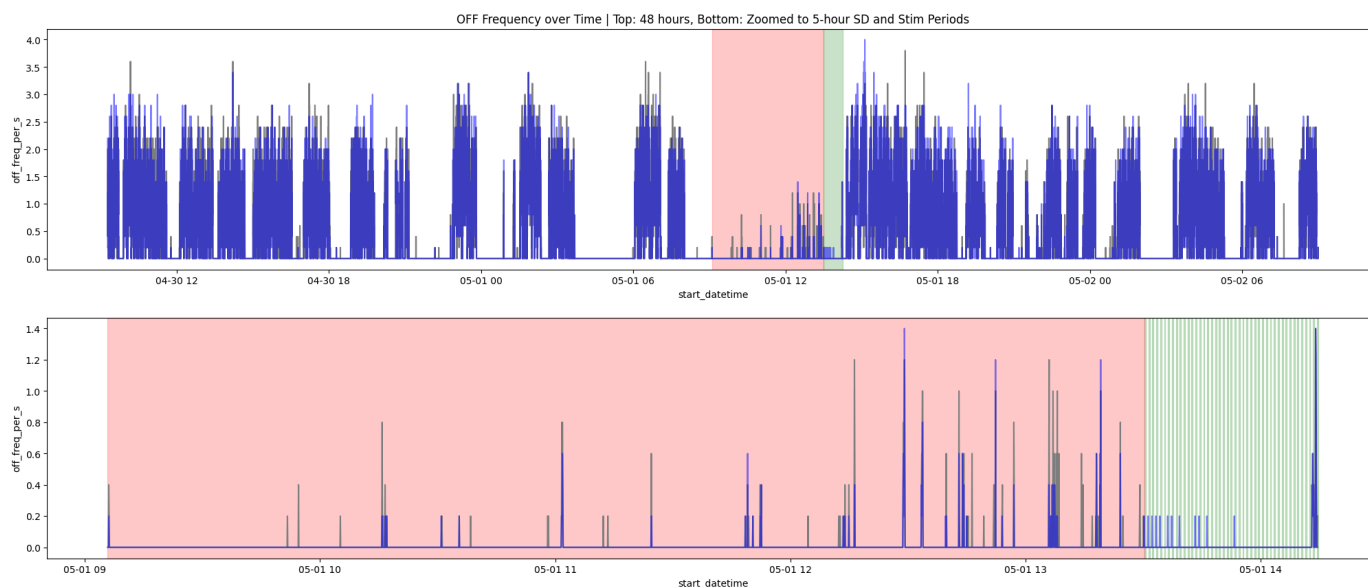
Experimental Design

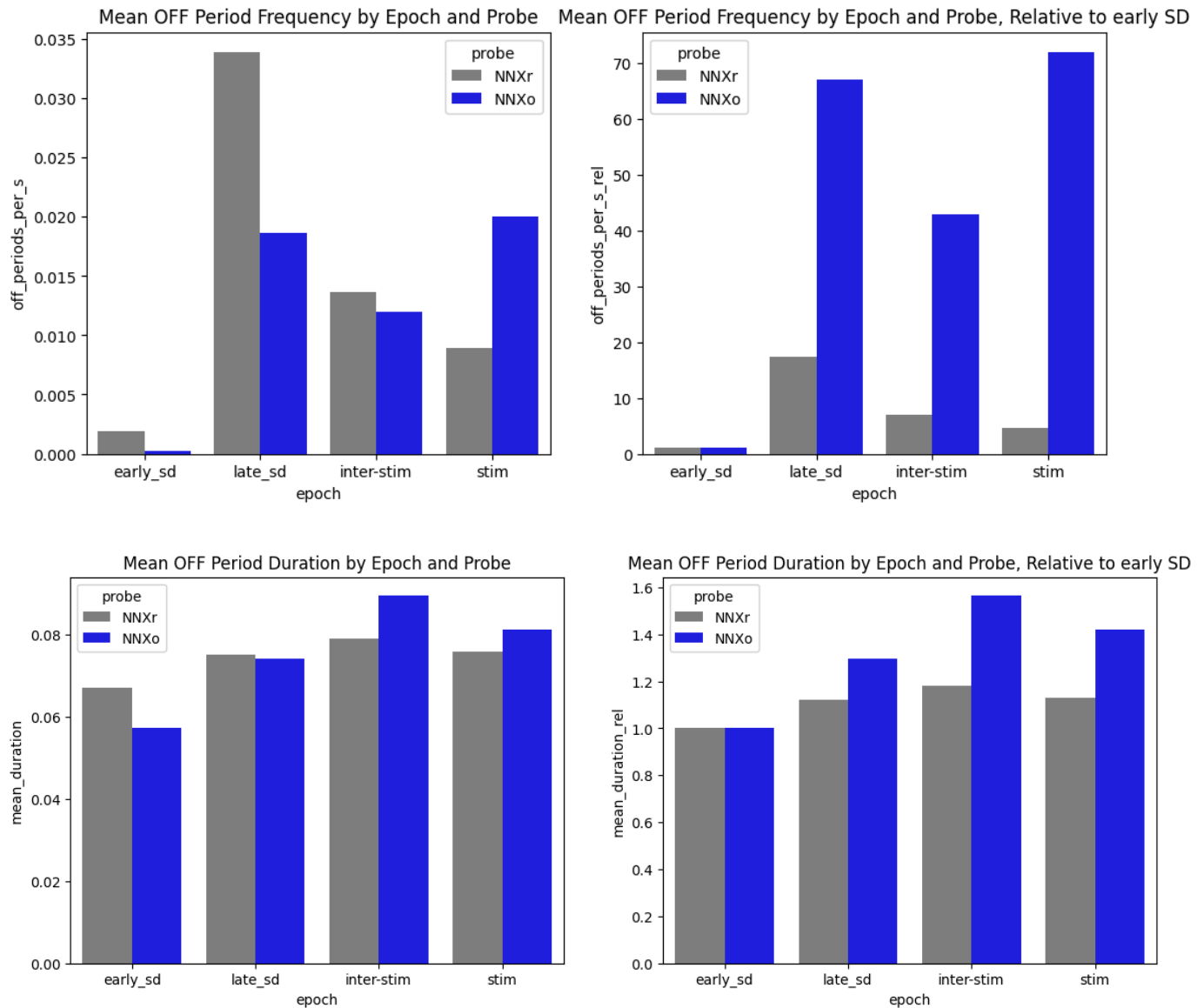
The design of the experiment was as follows:

- Mice had inhibitory ACR opsin expressed in SOM+ neurons.
- SD via novel objects from 9am to 2:15 pm
- From 1:30 - 2:15 (final 45 min of SD), laser was turned on for inhibition of SOM+ neurons. Laser was on for 20s, off for 40s. This process was repeated for the entire 45 min, resulting in 45 total stimulations.
- All of these analyses focus on comparing the stimulus period to the inter-stim period and preceding 4.5 hours of SD.

Whole-Probe OFF Period Detection

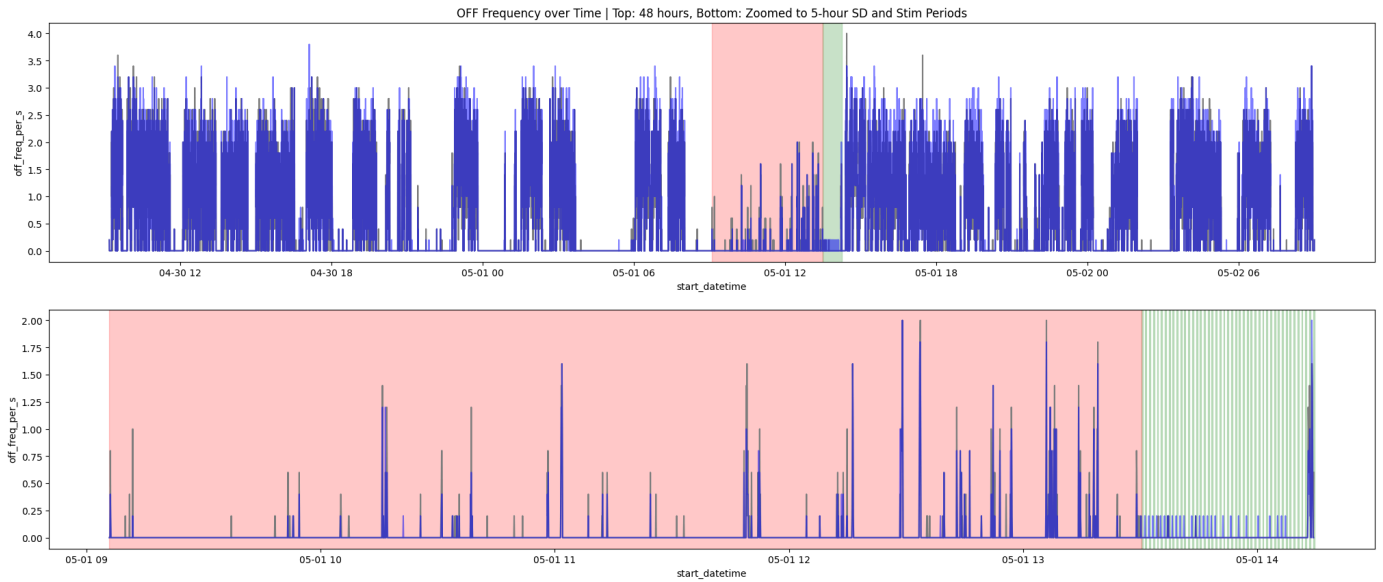
Here we use the canonical OFF period definition, where there must be at least 50ms of TOTAL MUA silence (i.e. no spikes on any channel) to be considered an OFF period:



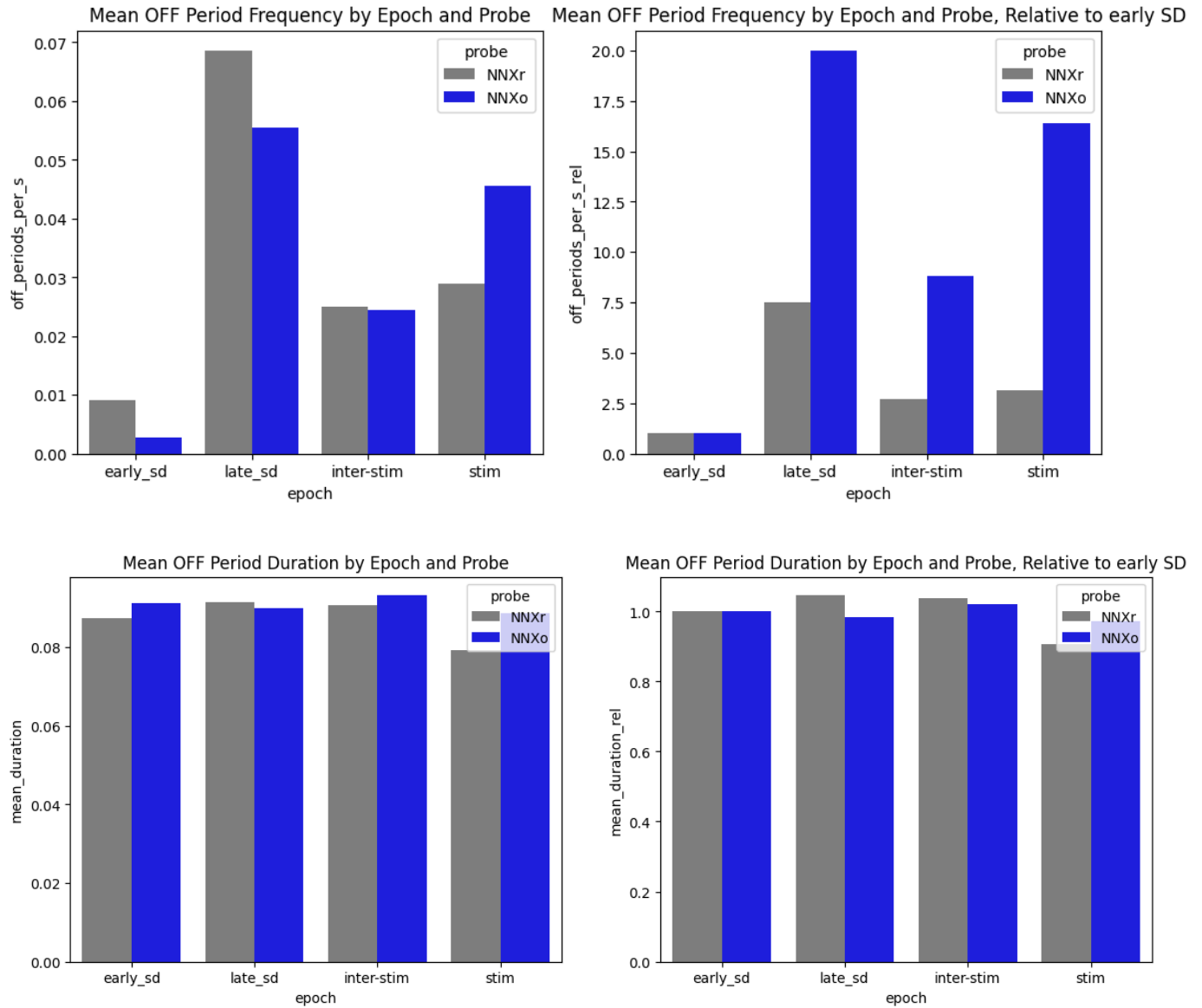


Hybrid OFF Period Detection

Here we first detect OFF periods at the single-channel level, and we then detect global OFF periods as those periods where some threshold number of single channels (12/16) are simultaneously in an OFF period. This is the exact definition used in the nature neuroscience paper.



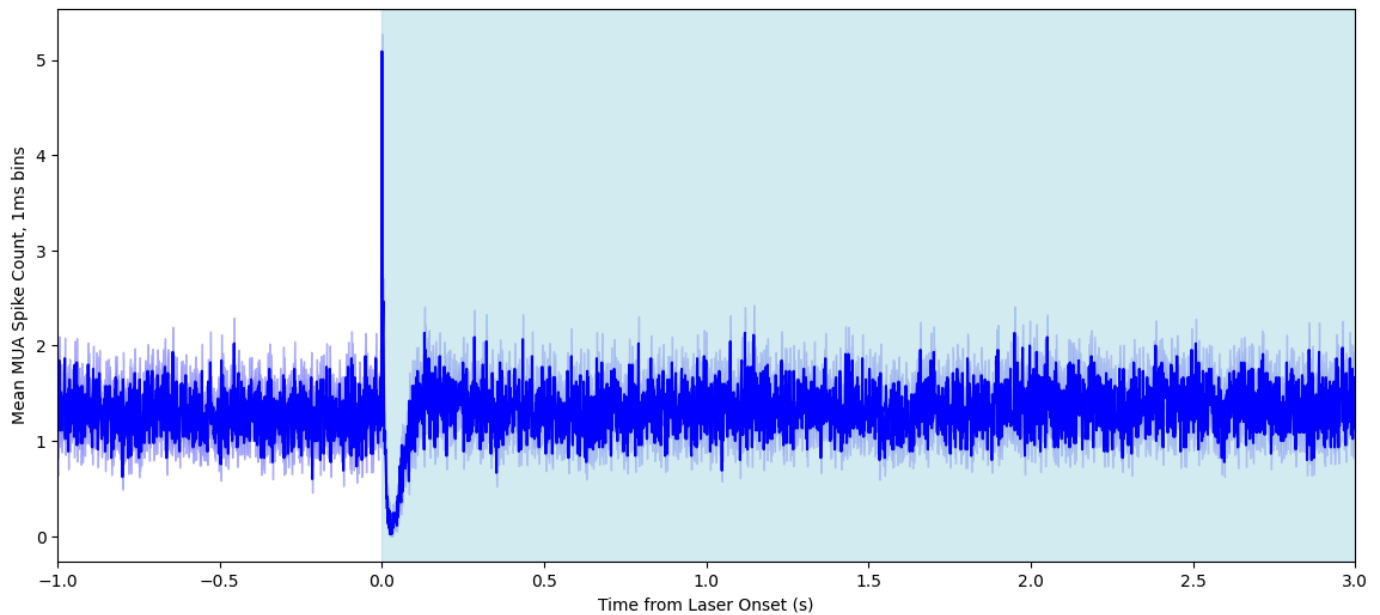
results look basically similar to the previous method:



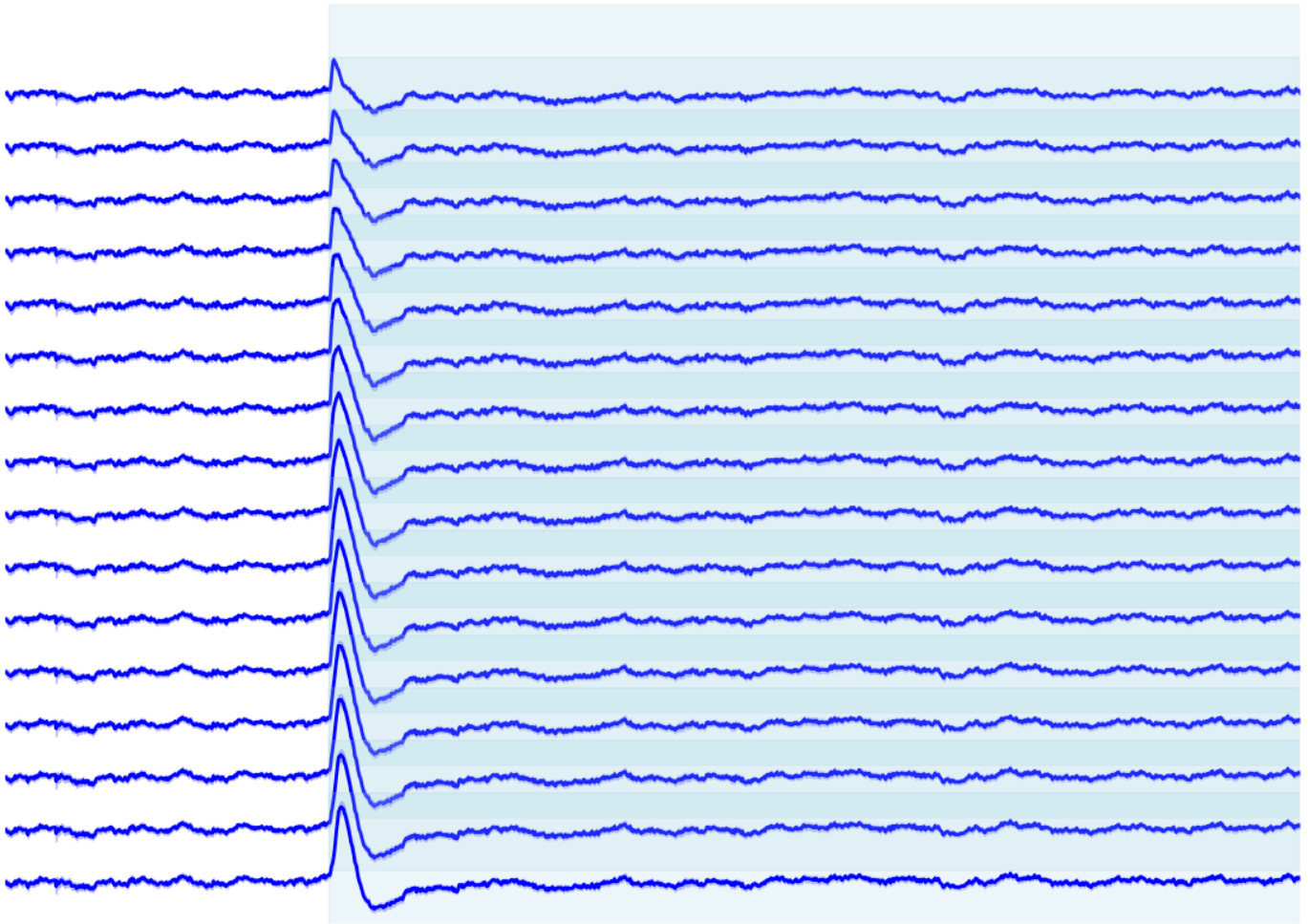
Are OFFs on the optrode being induced by the pulse itself?

Averaging together all pulses, two things are clear (shown in the below two plots respectively):

1. There is a sharp transient decrease in MUA activity immediately following the onset of the laser pulse (i.e. an OFF period)
2. There is clear positive deflection in the LFP immediately following the onset of the laser pulse (i.e. a slow wave)

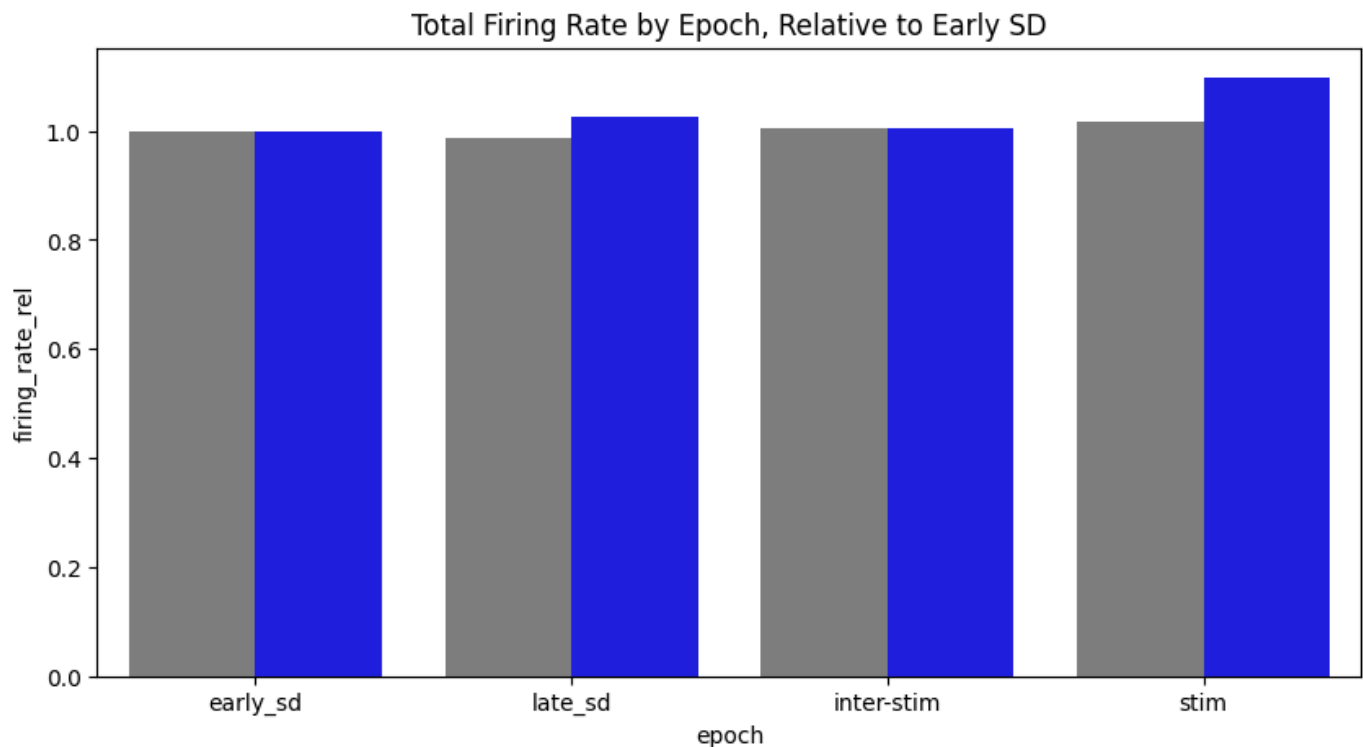


LFP, same exact timescale as MUA above:

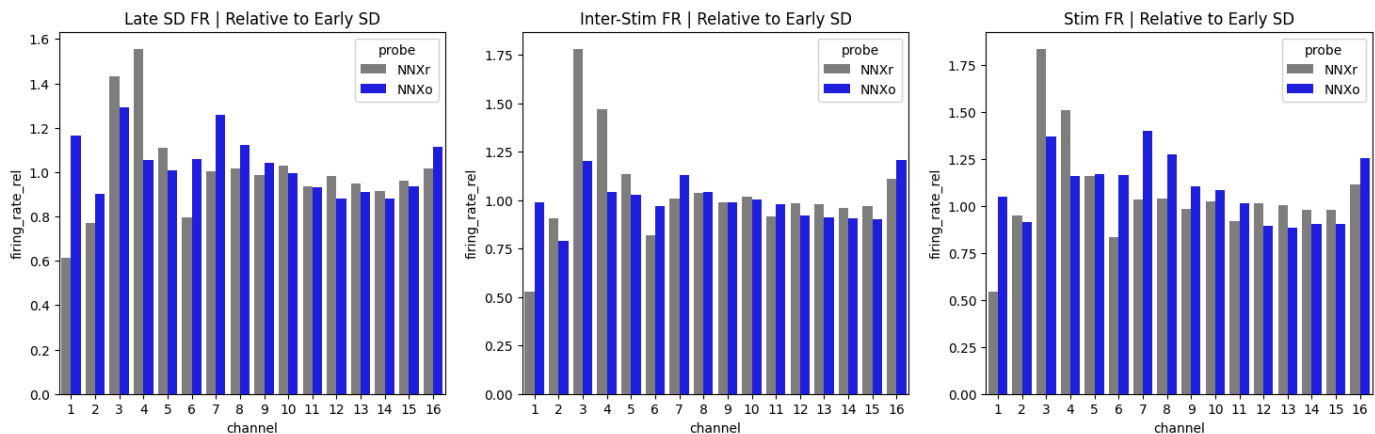


Overall Firing Rate, stim vs interstimulus interval

OVERALL firing rate during SOM+ inhibition does appear to be slightly increased on the optrode relative to contralateral control, and relative to the inter-stimulus period on the optrode:



Breakdown by channel:



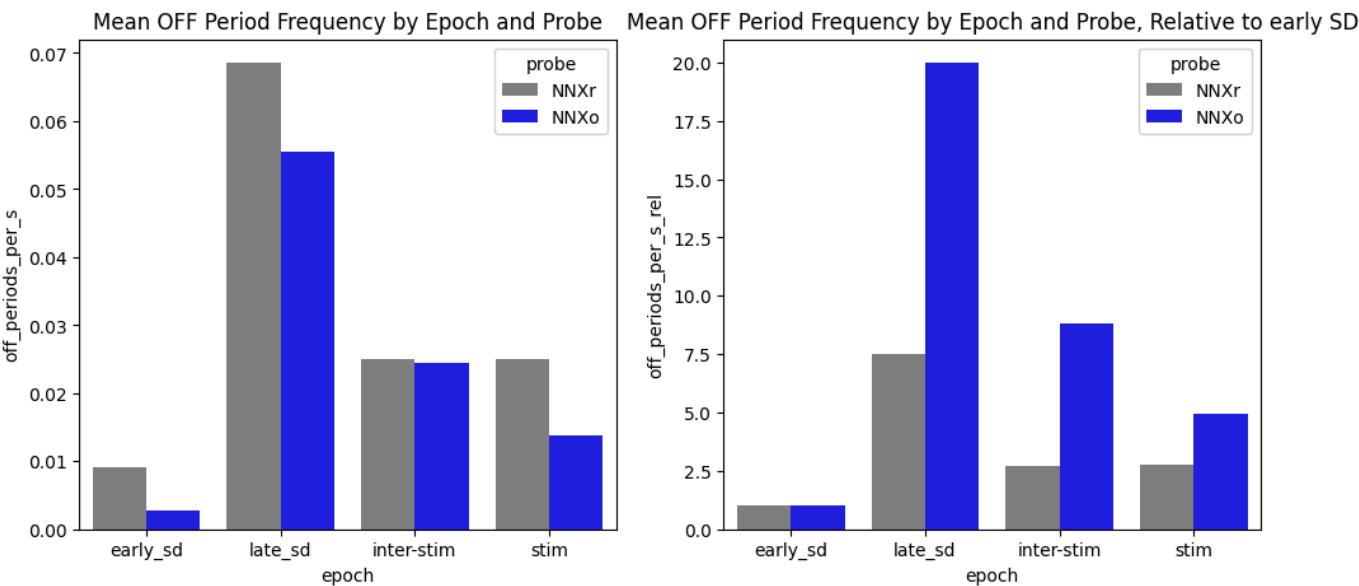
What if we remove the initial OFF periods associated with each laser-ON transient?

To remove the effect of the laser pulse itself, here we exclude from the 'stim' condition the first 500ms following the onset of each laser pulse.

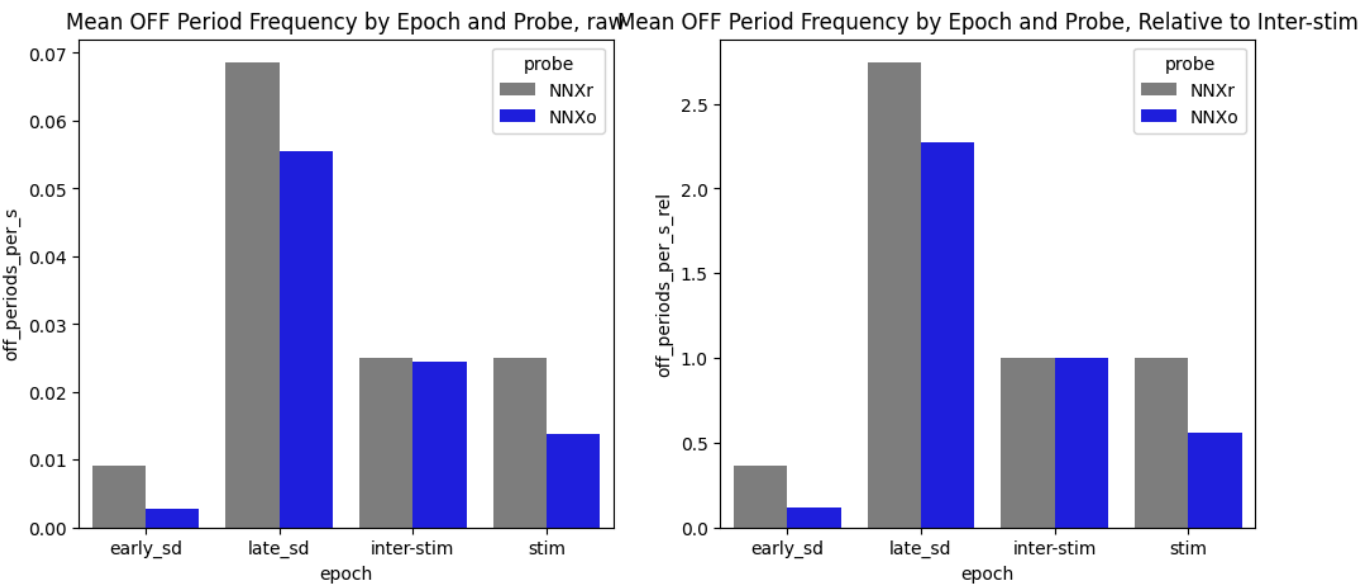
When expressed relative to the early SD period, there is still an increase in OFF period frequency on the Optrode compared to contralateral control (below, right), as in all analyses above.

However, perhaps the more sensible comparison is to compare the stimulus period to the inter-stimulus interval, since these are directly adjacent and thus more likely to be comparable in terms of

overall sleep pressure.



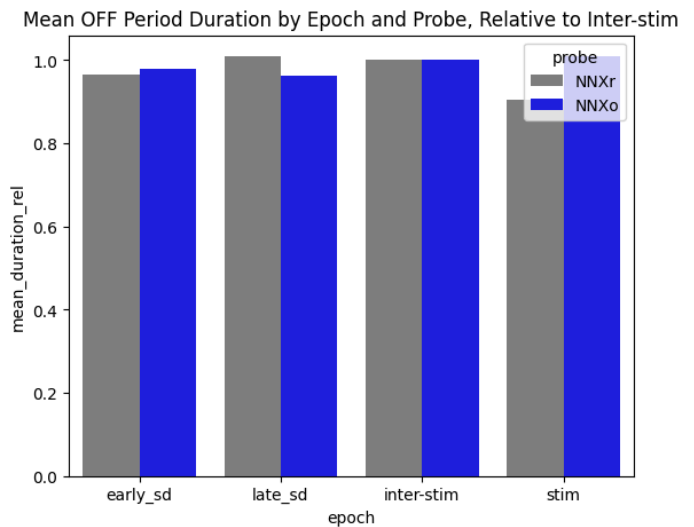
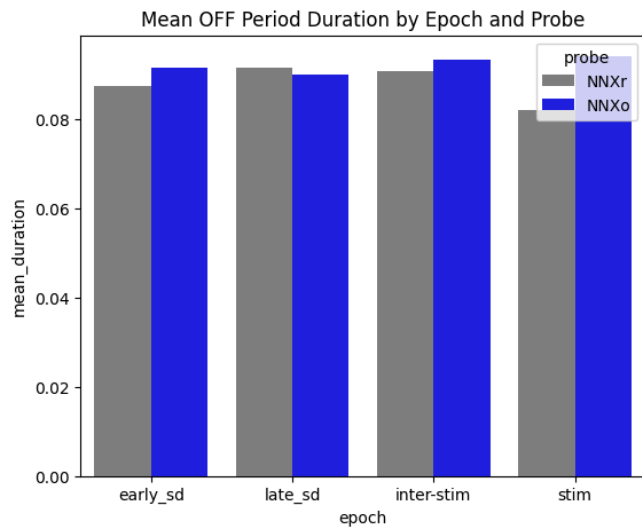
Relativizing the data to the inter-stimulus (laser-OFF) period:



Here we now see what looks like a decrease in off period frequency on the optrode relative to the contralateral control probe during the stimulus period, AND a decrease in off pperiod frequency on the optrode during the stimulus period relative to its own inter-stimulus interval period (perhaps the more relevant comparison).

This also begs the question of whether, at least for this type of experiment, we really need both both implanted.

Counter-intuitively, during SOM+ inhibition, there also appears to be a slight increase in OFF period duration on the optrode relative to the contralateral control probe, though not relative to the optrode's own inter-stimulus interval:



Conclusions

When we detect OFF periods just as I did in the Nat Neuro paper, and we remove the initial 500ms after each laser-on pulse (which very confusingly appears to be associated with a positive deflection in the LFP and a strong decrease in total MUA activity), I think we can say the following things based on these analyses with some confidence (note that these statements assume that laser-on actually does achieve SOM+ inhibition, which is not directly verified here):

1. Inhibition of SOM+ is associated with a decrease in OFF period frequency on the optrode compared to the inter-stimulus interval on the optrode when the laser is off.
2. Inhibition of SOM+ is associated with a decrease in OFF period frequency on the optrode compared to the contralateral control probe during the stimulus period, when both are referenced to their own inter-stimulus intervals. (i.e. it is not the case that the stimulus period is globally associated with a decrease in OFF period frequency - it occurs specifically on the optrode).
3. The confidence in the above two statements is slightly limited by the fact that the optrode OFF period frequency is already lower than contralateral control side during late SD relative to the inter-stimulus period. Nonetheless, I think that the best comparison is between the stimulus and inter-stimulus periods.